Personal Profile

Name: **Conor Glackin**
Occupation: Data Scientist | Computational Biologist | Machine Learning Specialist
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Work Authorisation: EU citizen (Irish), open to international relocation

Professional Profile

Data scientist with a PhD in statistics-driven environmental microbiology and a strong foundation in mathematics and statistics. Accomplished programmer (R, JMP, Python) with over 6 years' experience, adept at analysing complex real-world datasets, developing end-to-end machine learning models (Random Forest, regression, gradient boosting), and performing time-series analysis (LSTM using TensorFlow). Led projects involving predictive modelling, data visualisation, and data pipeline development in multidisciplinary scientific and technical teams. Effective communicator experienced in presenting complex quantitative findings to diverse audiences.

Key Achievements

- PhD on AI-driven prediction and forecasting of pathogenic bacteria in the Baltic Sea, creating an early warning system
- First-author manuscript on machine learning-based forecasting of bacterial presence published in Water Research (top ~2% journal); additional peer-reviewed publications using machine learning and statistics available (Google Scholar: https://scholar.google.com/citations?hl=en&user=8TSwVHYAAAA)
- Led process improvement and automation projects as a Data Scientist in a regulated pharmaceutical manufacturing environment

Technical Skills

- **Programming:** R, JMP/JSL, Python, SQL, Bash, HTML. Experience working in Linux environments
- **Machine Learning:** Random Forest, LSTM (TensorFlow), regression, gradient boosting, predictive modelling, time-series analysis
- **Statistical Methods:** Statistical modelling and validation, design of experiments (DoE), sampling plans
- **Data Visualisation:** R Shiny, ggplot2, Power BI
- **Data Engineering & Reporting:** Data workflows, data preparation, automated reporting pipelines
- **Bioinformatics & Tools:** MATLAB, Minitab, GitHub, NGS workflows (RNA-seq, ChIP-seq), BWA, GATK, SAMtools, ANNOVAR

Educational and Work Experience

- **Feb. 2022 - Feb 2026 PhD researcher, Leibniz Institute for Baltic Sea Research, Rostock, Germany**
  Thesis: *Quantification, Prediction and Forecasting of Vibrio vulnificus in the Southern Baltic Sea Using High-Resolution Data*
  - First author manuscript **"AI-Driven Forecasting of *Vibrio vulnificus* in the Southern Baltic Sea Using High-Resolution Data".**
    Independently conceived, designed, and authored a manuscript on machine learning-based forecasting of *Vibrio vulnificus* (a bacterium responsible for numerous deaths through infection each year), developing end-to-end workflows and models (Random Forest, LSTM using TensorFlow) for early-warning prediction; published in *Water Research* (top ~2% journal)
  - Led end-to-end data workflows including data preparation, feature engineering, model development, validation, and evaluation (ROC-AUC, AUPRC) on high-resolution environmental datasets
  - Designed and executed data-collection to create large-scale spatio-temporal dataset (15 locations, ~1500 samples), including data collection, processing, and analysis
  - Presented results at international conferences and collaborated with multidisciplinary scientific teams

- **Nov. 2019 - Jan. 2022 Data Scientist, Abbott Diabetes Care, Donegal, Ireland**

Role involved statistical analysis and programming projects including development and management of Power-BI dashboards to ad-hoc failure and root-cause analysis investigations, as well process optimisation, sampling plans, validation and document approvals. Presented analytical insights in daily cross-functional meetings, supporting real-time decision-making.

- Led development of the Smart New Analytics Project (SNAP), an internal analytics interface used across multiple teams for reviewing data and reports, building automated data pipelines and reproducible reporting workflows
- Led €30,000 SQL data warehouse upgrade, collaborating with the internal team and external vendors to implement improved data infrastructure
- Automated reporting workflows producing graphs and reports from product testing data (Excel files), reducing manual processing time from 8 hours to 2 minutes
- Led Lean Six Sigma Process Improvement project on predictive modelling for on-site blood glucose strip testing using upstream variables as predictors
- Conducted comparative statistical analysis of clinical and synthetic data for glucose conversion factors, supporting validation and quality processes

- **Sep. 2018 - Apr. 2019 Teacher, University of Galway, Ireland**
  Teaching classes of mathematics, applied mathematics, and statistics to university students in tutorials

- **Sep. 2018 - Jul. 2019 Tutor, SUMS: Support for Undergraduate Mathematics and Statistics, University of Galway, Ireland**
  Selected as a Statistics and Probability specialist.
  Worked on a one-to-one basis to assist university students with problems they cannot solve.

Bachelors and Masters Qualifications

- **Sep. 2018 – Aug. 2019 MSc. in Computational Genomics, NUI Galway, Ireland.**
  Thesis: *Predicting Links in Gene Regulatory Networks from other Networks*
  Worked in R-Studio using Logistic Regression, Lasso, Ridge, Elastic Net, SVM, and Decision Forest, with a 10×10- fold cross-validation used for analysing and comparing these methods.

- **Sep. 2014 – May. 2018 BSc. in Mathematics, NUI Galway, Ireland.**
  Thesis: *Determinants of Wartime Success*
  Project completed using Minitab and MATLAB

Additional Certifications & Skills

- Lean Six Sigma Yellow Belt [Abbott Diabetes Care, 2021]
- Advanced statistical software training (JMP)
- Advanced Microsoft Office proficiency (Excel, PowerPoint, Word)
- A2 Level German proficiency
- Full Driving Licence